FAISS in Practice: Recall–QPS Trade-offs, Query Hardness, and Robustness

Vasiliki Pantelopoulou & Panagiota Nalmpanti

https://github.com/Valia-p/Vector-Search-FAISS

ARISTOTLE UNIVERSITY OF THESSALONIKI

SCHOOL OF INFORMATICS
Data & Web Science MSc Program

Motivation

Approximate nearest-neighbor (ANN) search must balance accuracy and speed. Different FAISS methods make different trade-offs, but their behavior also depends on the dataset and query difficulty. In this project, we compare **IVFPQ**, **HNSWSQ**, and **LSH** on **SIFT**, **GIST**, and **GloVe**, and we study not only Recall@10 and QPS, but also **query hardness** and **robustness** to **noisy queries**.

Methodology

Datasets and Setup

We evaluate SIFT, GIST, and GloVe under a unified FAISS benchmarking pipeline. SIFT and GIST are treated as L2 datasets, while GloVe is handled as an angular dataset via L2-normalization and inner product.

- Exact ground truth: **FAISS Flat Indices**
- Main metrics: **Recall@10** and **QPS**

Methods

- **IVFPQ**: inverted file + product quantization
- **HNSWSQ**: graph search + scalar quantization
- **LSH**: hashing-based baseline

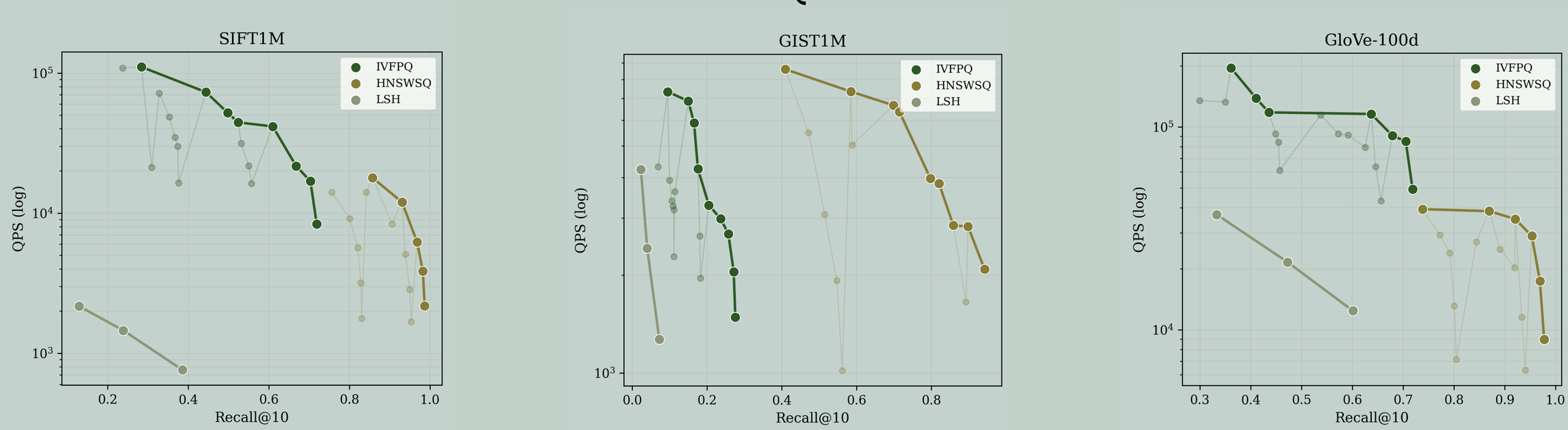
These methods represent different trade-offs between **compression**, **search effort**, **accuracy**, and **throughput**.

Experimental Setup

- Parameter sweeps: **nprobe**, **efSearch/SQ precision**, **nbits**
- Additional analyses: **RC**, **easy/hard queries**, **noise**
- Extra study: **adaptive effort on hard queries**

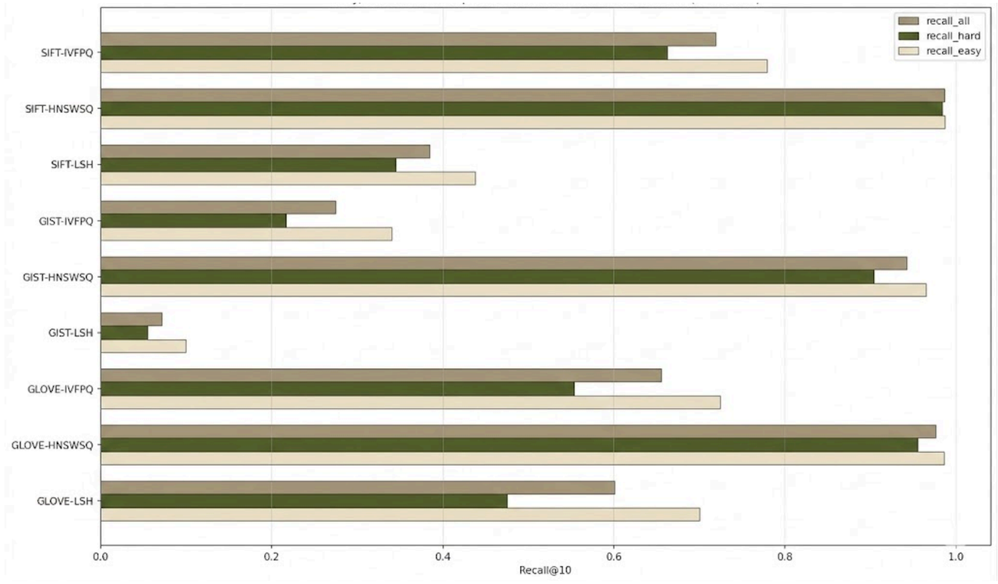
This setup captures both **accuracy–speed trade-offs** and **query difficulty effects**.

Recall and QPS Trade-off



No single winner. **Pick by priority, not by dataset** | **Recall-first**: **HNSWSQ** (0.94–0.99) | **Throughput-first**: **IVFPQ** | HNSWSQ–IVFPQ gap widens on GIST. **HNSWSQ's recall costs 10–40× more memory** | **HNSWSQ**: **~320–380 MB** (SIFT) | up to **~1175 MB** (GIST) **IVFPQ**: **~25–40 MB** for comparable-or-lower recall | **Memory-bound? Pick IVFPQ**

Query Hardness (RC)

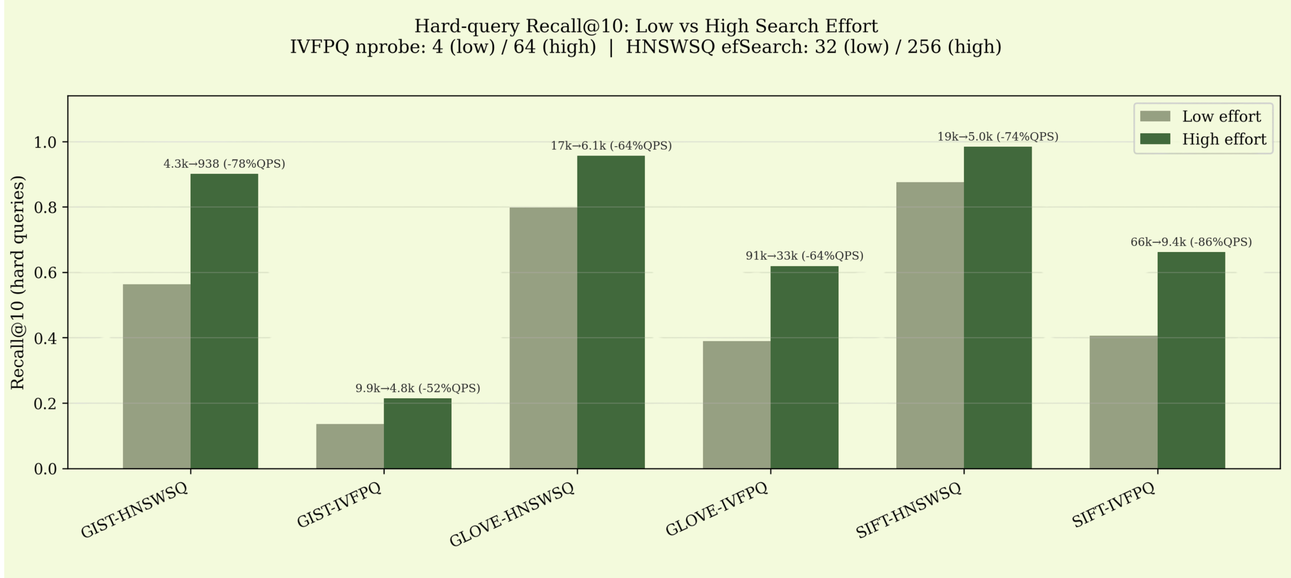


RC predicts difficulty & the easy-to-hard gap measures robustness
recall_easy > recall_all > recall_hard
RC is a valid hardness signal (low RC = hard)

Smallest gap = most robust:

• **HNSWSQ barely drops** on hard queries
• **IVFPQ drops more** & **LSH** stays **low throughout**

Adaptive Search



Adaptive effort recovers hard-query recall at a throughput cost.
• Raising nprobe/efSearch lifts hard-query recall on every method and dataset, but the gain isn't free:
QPS drops 52–86%: **• Best return on HNSWSQ** with **0.90–0.98 recall**
• Poor return on IVFPQ where **SIFT spends 86% QPS to reach only 0.66**

Takeaway: no single winner · HNSWSQ recall-first · IVFPQ throughput-first · RC predicts difficulty